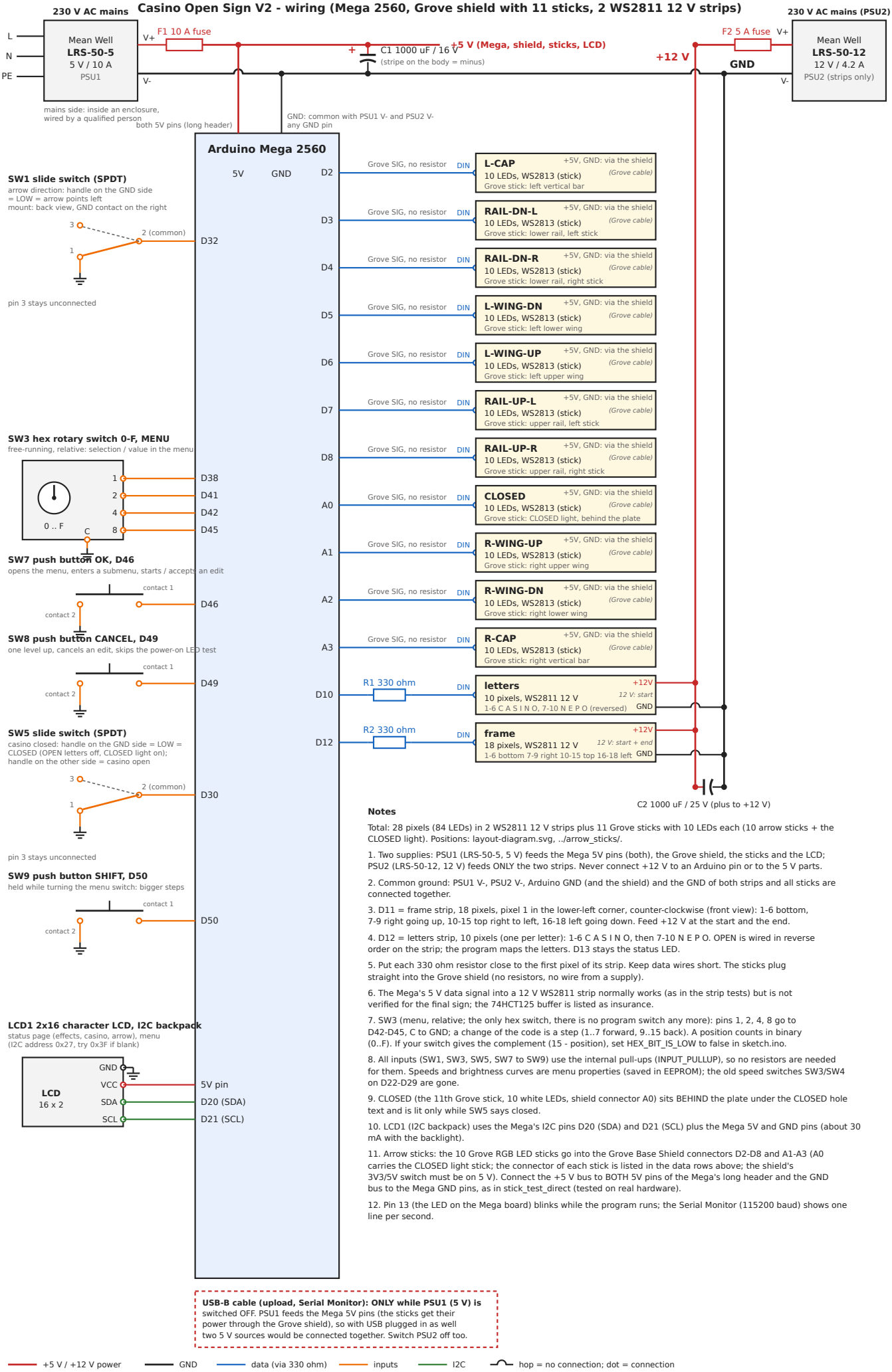

Casino Open Sign V2 - wiring (Mega 2560, Grove shield with 11 sticks, 2 WS2811 12 V strips)
230 V AC mains
L
N
PE
Mean Well
LRS-50-5
5 V / 10 A
PSU1
V+
V-
F1 10 A fuse
+5 V (Mega, shield, sticks, LCD)
C1 1000 uF / 16 V
(stripe on the body = minus)
mains side: inside an enclosure,
wired by a qualified person
both 5V pins (long header)
GND: common with PSU1 V- and PSU2 V-
any GND pin
230 V AC mains (PSU2)
F2 5 A fuse
+12 V
GND
Mean Well
LRS-50-12
12 V / 4.2 A
PSU2 (strips only)
Arduino Mega 2560
5V
GND
SW1 slide switch (SPDT)
arrow direction: handle on the GND side
= LOW = arrow points left
mount: back view, GND contact on the right
3
2 (common)
1
D32
pin 3 stays unconnected
SW3 hex rotary switch 0-F, MENU
free-running, relative: selection / value in the menu
0 .. F
C
1
2
4
8
D38
D41
D42
D45
SW7 push button OK, D46
opens the menu, enters a submenu, starts / accepts an edit
contact 1
contact 2
D46
SW8 push button CANCEL, D49
one level up, cancels an edit, skips the power-on LED test
D49
SW5 slide switch (SPDT)
casino closed: handle on the GND side = LOW =
CLOSED (OPEN letters off, CLOSED light on);
handle on the other side = casino open
D30
pin 3 stays unconnected
SW9 push button SHIFT, D50
held while turning the menu switch: bigger steps
D50
LCD1 2x16 character LCD, I2C backpack
status page (effects, casino, arrow), menu
(I2C address 0x27, try 0x3F if blank)
LCD
16 x 2
GND
VCC
SDA
SCL
5V pin
D20 (SDA)
D21 (SCL)
D2
D3
D4
D5
D6
D7
D8
A0
A1
A2
A3
D10
D12
Grove SIG, no resistor
DIN
L-CAP
10 LEDs, WS2813 (stick)
Grove stick: left vertical bar
RAIL-DN-L
Grove stick: lower rail, left stick
RAIL-DN-R
Grove stick: lower rail, right stick
L-WING-DN
Grove stick: left lower wing
L-WING-UP
Grove stick: left upper wing
RAIL-UP-L
Grove stick: upper rail, left stick
RAIL-UP-R
Grove stick: upper rail, right stick
CLOSED
Grove stick: CLOSED light, behind the plate
R-WING-UP
Grove stick: right upper wing
R-WING-DN
Grove stick: right lower wing
R-CAP
Grove stick: right vertical bar
+5V, GND: via the shield
(Grove cable)
R1 330 ohm
R2 330 ohm
letters
10 pixels, WS2811 12 V
1-6 C A S I N O, 7-10 N E P O (reversed)
+12V
12 V: start
GND
frame
18 pixels, WS2811 12 V
1-6 bottom 7-9 right 10-15 top 16-18 left
12 V: start + end
C2 1000 uF / 25 V (plus to +12 V)
Notes
Total: 28 pixels (84 LEDs) in 2 WS2811 12 V strips plus 11 Grove sticks with 10 LEDs each (10 arrow sticks + the CLOSED light). Positions: layout-diagram.svg, ../arrow_sticks/.
1. Two supplies: PSU1 (LRS-50-5, 5 V) feeds the Mega 5V pins (both), the Grove shield, the sticks and the LCD; PSU2 (LRS-50-12, 12 V) feeds ONLY the two strips. Never connect +12 V to an Arduino pin or to the 5 V parts.
2. Common ground: PSU1 V-, PSU2 V-, Arduino GND (and the shield) and the GND of both strips and all sticks are connected together.
3. D11 = frame strip, 18 pixels, pixel 1 in the lower-left corner, counter-clockwise (front view): 1-6 bottom, 7-9 right going up, 10-15 top right to left, 16-18 left going down. Feed +12 V at the start and the end.
4. D12 = letters strip, 10 pixels (one per letter): 1-6 C A S I N O, then 7-10 N E P O. OPEN is wired in reverse order on the strip; the program maps the letters. D13 stays the status LED.
5. Put each 330 ohm resistor close to the first pixel of its strip. Keep data wires short. The sticks plug straight into the Grove shield (no resistors, no wire from a supply).
6. The Mega's 5 V data signal into a 12 V WS2811 strip normally works (as in the strip tests) but is not verified for the final sign; the 74HCT125 buffer is listed as insurance.
7. SW3 (menu, relative; the only hex switch, there is no program switch any more): pins 1, 2, 4, 8 go to D42-D45, C to GND; a change of the code is a step (1..7 forward, 9..15 back). A position counts in binary (0..F). If your switch gives the complement (15 - position), set HEX_BIT_IS_LOW to false in sketch.ino.
8. All inputs (SW1, SW3, SW5, SW7 to SW9) use the internal pull-ups (INPUT_PULLUP), so no resistors are needed for them. Speeds and brightness curves are menu properties (saved in EEPROM); the old speed switches SW3/SW4 on D22-D29 are gone.
9. CLOSED (the 11th Grove stick, 10 white LEDs, shield connector A0) sits BEHIND the plate under the CLOSED hole text and is lit only while SW5 says closed.
10. LCD1 (I2C backpack) uses the Mega's I2C pins D20 (SDA) and D21 (SCL) plus the Mega 5V and GND pins (about 30 mA with the backlight).
11. Arrow sticks: the 10 Grove RGB LED sticks go into the Grove Base Shield connectors D2-D8 and A1-A3 (A0 carries the CLOSED light stick; the connector of each stick is listed in the data rows above; the shield's 3V3/5V switch must be on 5 V). Connect the +5 V bus to BOTH 5V pins of the Mega's long header and the GND bus to the Mega GND pins, as in stick_test_direct (tested on real hardware).
12. Pin 13 (the LED on the Mega board) blinks while the program runs; the Serial Monitor (115200 baud) shows one line per second.
USB-B cable (upload, Serial Monitor): ONLY while PSU1 (5 V) is switched OFF. PSU1 feeds the Mega 5V pins (the sticks get their power through the Grove shield), so with USB plugged in as well two 5 V sources would be connected together. Switch PSU2 off too.
+5 V / +12 V power
GND
data (via 330 ohm)
inputs
I2C
hop = no connection; dot = connection